Kirinyaga University

Course Code & Title: SSE 2302 COMPONENT BASED SOFTWARE ENGINEERING SYLLABUS

Target Audience: BSc Software Engineering, Year 3 Sem 1

Credit Hours / Contact Hours: 3 Credits / 45 Contact Hours

Pre-Requisite: Software Engineering

Lecturer: Samuel Maina

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Purpose

This course introduces the principles, concepts, methodologies, and practices of Component-Based Software Engineering (CBSE). Learners examine the identification, specification, development, selection, reuse, integration, testing, deployment, and maintenance of software components. The course explores component-based architectures, interfaces, component composition, software reuse, middleware, APIs, web services, and modern component technologies. Learners develop the knowledge and practical skills required to design, integrate, and manage reliable, reusable, and maintainable component-based software systems.

Course Learning Outcomes

Upon successful completion of this course, the learner should be able to:

1. Analyze the principles, concepts, characteristics, and benefits of Component-Based Software Engineering and its application in modern software development.
2. Evaluate software components based on their interfaces, functionality, quality attributes, compatibility, reusability, and suitability for integration into software systems.
3. Apply component-based software engineering practices to identify, specify, design, compose, integrate, test, and deploy reusable software components.
4. Design component-based software architectures using appropriate component models, interfaces, connectors, APIs, middleware, and integration techniques.
5. Analyze challenges related to component security, interoperability, dependency management, versioning, maintenance, and evolution of component-based systems.
6. Develop a component-based software solution that demonstrates effective component selection, reuse, integration, testing, deployment, and documentation.

Contact Hours: 45 Hours

Course Content and Delivery Plan

WEEK	TOPIC	SUB-TOPICS	INTENDED LEARNING OUTCOMES	LEARNING EXPERIENCES
1	Introduction to CBSE	Concepts, evolution, benefits and challenges	Define and explain CBSE	Lecture, discussion
2	Software Components	Characteristics, interfaces and contracts	Identify and explain components	Lecture, exercises
3	Component Interfaces	Interface specification, contracts and compatibility	Specify and analyze interfaces	Practical, case study
4	Component Architecture	Architecture, connectors, dependencies and patterns	Design basic component architecture	Lecture, design exercise
5	Component Identification	Requirements, domain analysis and specification	Identify and specify components	Group work, case study
6	Component Reuse	Reusable components, repositories and reuse strategies	Select and evaluate reusable components	Practical, discussion
7	Component Composition	Integration, adapters, wrappers and compatibility	Integrate and analyze components	Practical, group project
8	Middleware and APIs	Middleware, APIs, web services and REST	Explain and use component technologies	Demonstration, practical
9	CBSE Development Process	Lifecycle, design, implementation and deployment	Apply CBSE development processes	Lecture, project work
10	Component Testing	Unit, interface, integration and regression testing	Develop and evaluate test cases	Testing practical

11	Security and Maintenance	Security, versioning, maintenance and evolution	Analyze and apply security measures	Case study, practical
12	Deployment and Management	Deployment, configuration, dependencies and repositories	Deploy and manage components	Demonstration, practical
13	Project Presentation and Review	System demonstration, testing and evaluation	Evaluate and present a CBSE project	Presentations, assessment

Assessment	**Weight (%)**
Continuous Assessment (Assignments, Practical Labs, Research Project, Presentations)	**30**
Final Examination	**70**
Total	**100**

Core Reading Materials

1. Sommerville, I. **Software Engineering** (11th Edition). Pearson Education.
2. Pressman, R. S., & Maxim, B. R. **Software Engineering: A Practitioner's Approach** (9th Edition). McGraw-Hill.
3. Szyperski, C., Gruntz, D., & Murer, S. **Component Software: Beyond Object-Oriented Programming** (2nd Edition). Addison-Wesley.
4. Heineman, G. T., & Councill, W. T. **Component-Based Software Engineering: Putting the Pieces Together.** Addison-Wesley.

Recommended References

1. Brown, A. W., & Wallnau, K. C. **The Current State of CBSE.** IEEE Software.
2. Crnkovic, I., & Larsson, M. **Building Reliable Component-Based Software Systems.** Artech House.
3. Cheesman, J., & Daniels, J. **UML Components: A Simple Process for Specifying Component-Based Software.** Addison-Wesley.
4. Fowler, M. **Patterns of Enterprise Application Architecture.** Addison-Wesley.

5. Gamma, E., Helm, R., Johnson, R., & Vlissides, J. **Design Patterns: Elements of Reusable Object-Oriented Software.** Addison-Wesley.
6. Newman, S. **Building Microservices** (2nd Edition). O'Reilly Media.
7. Richardson, C. **Microservices Patterns: With Examples in Java.** Manning Publications.